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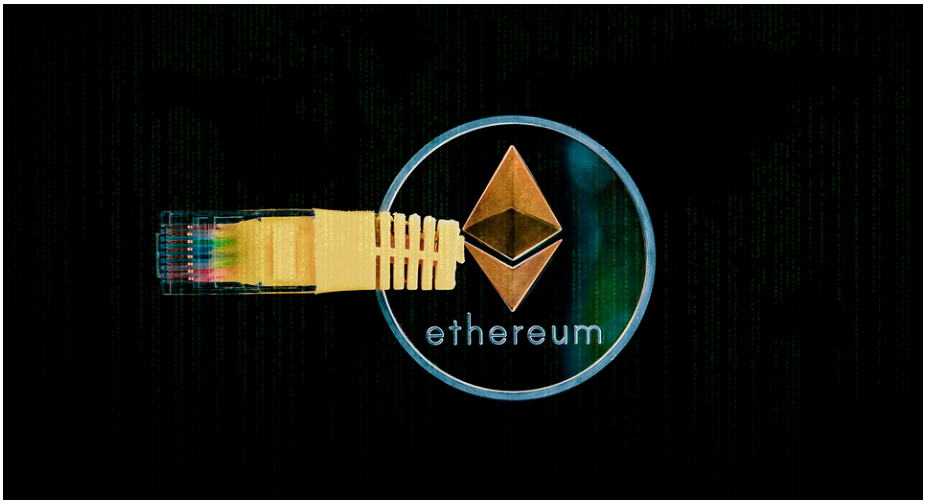
How to Mine Ethereum

608 comments | CryptoCompare | 10 May 2019 | 2.24 M



If you're interested in dedicating computing resources to mining cryptocurrency for yourself, this article explains how the basics on how to begin mining Ethereum, and the different mining software options.

The Background

Ethereum mining has evolved considerably over the last few years. There's a drive towards greater and easier participation, and away from the realms of the tech-savvy early adopters. Complicated command-line mining tools have become easier to use, and streamlined. GUI mining software has sprung up that makes it really simple to get going and start earning Ether (or ETH), the currency of Ethereum.

Mining Ethereum works in a similar way to mining Bitcoin, and was designed for a similar reason. The mining process underpins the decentralization of the cryptocurrency by incentivizing computers to validate transactions on the network. Computers compete to be the one to validate transactions and are tasked with solving a complex computational equation in a process called proof-of-work. With Ethereum, whoever solves the equation first is rewarded with 2 ETH (recently reduced from 3 ETH) as well as all the transaction fees known as "gas."

As ETH increased in price in 2016 and 2017, so too did the Ethereum Hash Rate - a number that reflects how much processing power is collectively being used to solve the proof of work algorithm. Despite a large decline in the price of Ethereum since early 2018, the hash rate has still stayed high. This is partly thanks to increased participation, and partly because of the development of powerful dedicated Ethereum mining hardware in form of Application Specific Integrated Circuit (ASIC) systems from the likes of Bitmain, such as the Antminer E3.

Due to their massive potential hashing power, and the large corporations that build and run them, ASICs are considered a threat to the decentralization of the network. This has sparked debate among the Ethereum community regarding whether its proof of work algorithm should be changed to resist such technology - it's a big potential change to the estimated $655 million annual market for Ethereum's mining rewards.

For most of us, mining cryptocurrency will be an activity undertaken using GPUs – either the one built into your computer, or on a gaming card such as those from NVIDIA or AMD. Anyone can try it – but to be profitable, you need an up-to-date GPU, or a dedicated GPU mining rig.

But why should you make the effort to mine ETH?

You can easily buy Ether on cryptocurrency exchanges, but mining is different. Here are three reasons to give it a go:

- **It's an easy way to get ETH** – hobby mining is a low risk, low cost (at least to start with) way of obtaining cryptocurrency and getting involved in the crypto movement
- **You're a believer** – by mining you contribute to the stability of the network and become an active part of the community
- **It can subsidize the cost of a high-end GPU -** Although the profits aren't huge these days, it will help pay back that expensive gaming graphics card

How to get going and start mining Ethereum today

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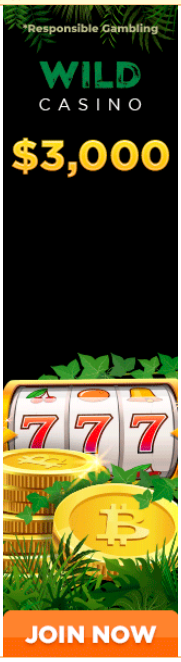

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These days, mining Ethereum is really accessible – a far cry from the command line interfaces of the past (although the old [illegible] still exist if you want to go deeper). Here's our three step guide to getting going:

1. **Set up your mining hardware**

Mining Ethereum, at its heart, requires a computer that contains the same elements as any computer you are already familiar with – a CPU, RAM, storage, Power Supply Unit (PSU), cooling fans. However, to be profitable, you will need a secondary processing location - a GPU or ASIC circuit. Most people getting started begin with a gaming graphics card such as the NVIDIA GeForce GTX 1080, although new models come out all the time.

To maximize your profits, ensure you have the latest software drivers for your hardware, and get an efficient PSU to power it. An efficient PSU will not only reduce costs, it will also allow you to overclock your graphics card to get more performance from it. Also ensure you have efficient cooling in place, especially if you're overclocking your GPU.

The aim is to get the most performance for the least power - less power in + more processing = more profit. It's that simple.

Progressing from a single graphics card in your desktop PC, you might consider a mining rig – these are machines specially designed for one purpose only – to mine cryptocurrencies.

2. **Set up your Ethereum wallet**

Before you start mining on your mining hardware, you should set up a crypto wallet. Some mining software includes a built-in wallet, but the most secure way to hold your cryptocurrencies is in your own wallet (it's best to do your own research to understand why).

The simplest way is to get an Ethereum wallet is to use MyEtherWallet (MEW) [https://www.myetherwallet.com/], which is free and offers security and convenience. MyEtherWallet can be accessed via Browser using the company's own Metamask-powered browser extension.

3. **Set up your software**

Here's where you need to make a decision, do you want to use a dedicated command-line miner which is more complicated to set up but is more customizable, or do you want to try a GUI miner which is easy to get going but not quite as versatile?

Go manual on the Command Line

The three main options for manual mining are:

- Claymore's Dual Ethereum Miner [https://bitcointalk.org/index.php?topic=1433925.0]
- Phoenix Miner [https://bitcointalk.org/index.php?topic=2647654.0]
- EthMiner [https://github.com/ethereum-mining/ethminer]

All of them require setting up and editing of configuration files - which demands some knowledge of Command Line Interface.

Claymore has long been top of the list of Ethereum miners - respected for its profitability and the control it offers over the hardware tweaking and multi-coin mining capabilities.

PhoenixMiner is heralded by many as a more efficient system that also offers additional stability over Claymore.

EthMiner - a derivative of the extra techy Geth [https://github.com/ethereum/go-ethereum/wiki/geth] is a low resource code that can be used to mine any Ethash-based crypto, meaning it can also be dedicated to Ethereum Classic and Metaverse for example, by anyone looking to configure it. It will demand more in-depth work to get up-and-running, however.

All have dedicated communities, and various step-by-step guides to installation are easily found. Which option you go for will depend on your confidence level, and your willingness to get into the nuts and bolts of computing.

One additional complication - to use these miners you'll also need to join a mining pool. This is because mining Ethereum on your own - certainly with a single small rig - is a pretty much thankless task. The nature of Proof of Work and the current crypto ecosystem means that your chances of gaining any rewards are very low indeed – it's a lottery whether you'll ever be the one to get the block rewards.

But similar to lottery pools, you can join forces in a mining pool such as Ethpool.org, or similar, and dedicate your resources (and rewards) to a collective. This will give you a regular income depending on the amount of hash power you contribute, but you'll need to set up your miner to participate in your chosen pool.

Another option is to pick a platform that rolls mining and pooling into one package, for example Nanpool.org also offers its own Nanominer to anyone that wants to use it - though it can also be configured to work with Claymore and other software.

Keep it simple with a GUI miner

For those looking for a really easy time, look no further than Cudo Miner [https://www.cudominer.com/]. A user-friendly piece of software that offers a more familiar graphical interface for those looking to dedicate their GPU and CPU resources to mining.

Its simple set up will allow you to join its pool, and it mines a number of cryptocurrencies including Ethereum. A key feature of Cudo Miner is that it offers an auto-switching algorithm, allowing users to hand over mining choices to the software to ensure it mines the most profitable cryptocurrency. A neat feature is that you can still receive payment in Ethereum so this means it's a more profitable way to gain Ethereum through mining.

Cudo Miner comes in simple to install packages for Windows, MacOS and Linux – no knowledge of the command line is required to get going. So in terms of learning curve, it is by far the shallowest option we have come across thus far – it's click-and-go, you don't even need a wallet to begin with.

GUI mining is a great way to test the water, and see if you want to go deeper.

Conclusion

Ethereum Mining has changed immeasurably in recent years. When we wrote the original guide four years ago, it was a very complicated process. Even to create a wallet address required syncing the entire Ethereum blockchain.

Now it's so simple to start. If you've got a suitable graphics card and are interested in the space, then why not try mining today!

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Post

mineshop.eu
Just now

Missing the ETH mining , fun times.

reply Agree

sophiaanna506
10 months ago

Register now to reword 800 TRX inbox for more

reply Agree 1 1

mikear1432
1 year ago

lol what are all these advertisement comments. everyone sounds like a bot.

1 reply Agree 1

pigdude47
1 year ago

When I download this mining software it's just a zip file. What do I do with the file?

reply Agree

afitco1010
1 year ago

Here is how to cryptocurrency works ?

reply Agree

regitom431
2 years ago

The crypto mining space is winding up progressively more marvelous as higher calculation power is used in mining. On the other hand you can take a chance and make easy profit on bitcoin, you need to join Digitalcurrency14@gmail com Bitcoin mining pool to mine adequately and make a lot of profit, all they do is to connect your blockchain to there mining pool and that way you'll make a lot of profit like I did. I don't really know how they get this done but there service is beyond Imagination. Connect to there Mining pool today and get yourself a better earning on bitcoin!

1 reply Agree

regitom431
2 years ago

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reply Agree 1

hesafa2065
2 years ago

Times have changed. Technology is evolving and opportunity are coming on cryptocurrency everyday. There are experts across the globe who understands how to make profit fast on cryptocurrency luckily for me I was able to come across one few months ago and the result was overwhelmed. I contact him on his email digitalcurrency14 @gmailcom He explained the whole process to me and we both got a deal after the agreement. To my greatest surprised I made a lot of profit in few weeks which I wouldn't want to disclose here but I'm glad of his service

reply

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2 years ago

Work with Minergate! I optimize the work to get a good profit tg inga_drozdova

reply Agree

abrahameric11
2 years ago

I have been thinking the best and most lucrative option is Cloud Mining not until i came through (Bitcoin mining expert online). However finding an authentic and reliable Miner is a challenge. To my own understanding we should get the best solution because when you make lots of research about a business or an investment company you will get to obtain lots of information from different people on their experience with that particular business or investment from the deduction you made you can now decide what action to take. I receive 4.5BTC within 72hrs and a Minimum of 0.8 BTC a day and a weekly profit of 4.86BTC. He added me to his elite telegram group and i will advice you all seeing this review to contact him for more about mining right here.. BITCOINMININGEXPERT11 AT GMAIL DOT COM

reply Agree

latoyamann12
2 years ago

Honestly this is really amazing, my whole life i never believed in Bitcion mining until i decided to give it a try, So i contacted (MORGANJIMCRYPTONARY@GMAIL. COM) then i found out he was legit when i saw his profile on google, With no upfront payment, I just funded my wallet with 0.11BTC then within 24 hours it increased to 0.15BTC, you will see the increase in your bitcoin wallet within 24 hour. Please tell him from Latoya.

reply Agree

blqtnliso
2 years ago

Antminer S17+ 67TH/s

www.dragonxminer.com/product/antminer-s17-67th-s/

reply Agree

abrahameric11
2 years ago

My bitcoin was stolen and following is how it happened? I connect my blockchain wallet to the blockchain mining network with the help of a ant-miner hardware. He asked me to register a blockchain account and send him the wallet id and password so he can do the connections. I shared with him my user name and password information, After three days my bitcoin wallet was empty before i finally came across Queen Hellian and i really appreciate all what he did by getting my BTC back within few days. kindly get hold of him right here:
BITCOINMININGEXPERT11@GMAIL.COM

reply Agree

InvestmentHunter
2 years ago

First, thanks for the post. It was informative. I stopped trying to mine at home a year ago because A) it just is not even close to cost-effective to try to get BTC or ETH with the average price of electricity, and B) I discovered I was getting so much skimmed off the top that it was worth even less than expected by these easy-to-use systems.

For example, MinerGate was simple to run if you had compatible hardware, but it generated sooooo little that it took forever to get practically nothing from it. Furthermore, after some more detailed analysis, it seemed like what some of these tools and places were doing was making themselves money by getting volunteered CPU/GPU power and then throwing back a little bit to keep people interested, but you couldn't even make enough to transfer any of it! So, you have to keep running 24/7 to get the minimum amount to transfer it out, and that's when you see it's far better to invest money rather than to mine. Also, another consideration is that, when you end up giving up, you are leaving money in their pockets: they control the wallets so they can keep the abandoned coins. It's like that famous crime where the the programmer steals a few cents from every transaction: it doesn't seem like much so who would pay attention, but across millions of transactions, it's a crapload of money!

The power is really the issue. Unless you are living off of someone else paying the bills, even most cheap rates are not enough to break even with mining. The pools take a huge cut and doing it yourself will probably cost twice as much as you make. (I guess if you make a little and count of BTC being $250k per coin someday, maybe it's worth it, but if you believe it'll be $250k per coin, then you should just take that powerbill money and buy BTC instead!!!

With all that said, it you are mining some crapcoin that you get lucky with on timing for a sale, that's a different story I guess, but how do you pick a coin to go with? It's just gambling. RavenCoin, Monero, etc.... Who knows.... I've searched the Internet for advice and found nothing compelling.

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ground every day than I can make spending $30 or $40 extra on my electric bill to make a couple of dollars. (Again, assuming my coin of choice is ETH or BTC, and not some other coin that may go up or may crash to nothing.)

I've seen some online videos comparing Cudo Miner to others like BetterHash, which is interesting, but they usually have huge rigs and don't share what it costs them in electricity, they are just showing which is better on their rigs.

Anyway, thanks for the info.

What I'd really like to see is an informed, in-depth analysis of which AltCoin would be best to convert your Cudo Miner profits into, since there's a choice of BTC, ETH, Monero, RavenCoin, and some fiat. Which is better if you want to gamble on the AltCoin price of choice three years from now? Seems like RavenCoin at first glance since the current price is so low.

reply Agree 2

Korum
3 years ago

You can use myMiner tool for mining ETH, which uses ethMiner mentioned in the article.

reply Agree

kll52405
3 years ago

I am pretty new to these crytos and mining but I am very interested in learning more. I like the EOS coin

1 reply Agree

e7e126ab81
3 years ago

Hi, i am trymg to make this worck, but first, the step geth --rpc is take a few days.
The link for Ethminer is not working. I found Ethminer but is not an instaler, so i can´t continue.
Please, help-

reply Agree

d3414d95ce
3 years ago

Oh, I just found that your phone can also mine crypto at The Crypto Sight: thecryptosight.com/electroneum-creates-smartphone-that-mines-crypto/

reply Agree

ronbaer67
3 years ago

they must have changed the zip file because i upzipped it to my drive and there is no .exe file at all just folders and .go files

reply Agree

chimnedum.nwoke
4 years ago

Hello, I'm at step 7 and whenever I type "geth --rpc" it shows
Fatal: Error starting protocol stack: datadir already used by another process

pls I need help

reply Agree 1

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